

1. Direct Verdict: Did All Reports Give Correct Reports Based on Business Rules?

AUDIT VERDICT: COMPLIANCE RATE ONLY 38.46% (5 PASS, 8 DEFECTIVE)

NO. The reporting suite fails to provide accurate financial figures across 8 out of 13 modules.

While the core foundational banking/cash ledgers maintain strict mathematical parity (**Payment Accounts ৳ 54,100.00 $\equiv$ Cash Flow Net ৳ 54,100.00**), the secondary reporting modules suffer from severe SQL aggregation flaws, hardcoded operating expense omissions, inverted cumulative bounds, silent return deductions, and false deficit presentations.

13

TOTAL AUDITED

5

PASS (38.5%)

5

CRITICAL FAIL (38.5%)

3

PARTIAL / UI (23.0%)

CLOSED-LOOP FINANCIAL PARITY EQUATION:

$\Phi = \sum \text{Cash In (৳ 369,500.00)} - \sum \text{Cash Out (৳ 315,400.00)} - \sum \text{Payment Accounts (৳ 54,100.00)}$

$\Phi = \text{৳ } 54,100.00 - \text{৳ } 54,100.00 \equiv \text{৳ } 0.00 \text{ (ZERO VARIANCE CONFIRMED)}$

→ Core cash ledger is sound, but secondary reporting queries are desynchronized and defective.

2. Master Cross-Report Audit Matrix

Cross-referencing every live report module against underlying transactional sources of truth (Cash Flow, Payments, Accounts, Sales, Purchases, Expenses, Assets, Loans, Investments).

#	REPORT MODULE & PATH	REPORTED LIVE VALUES	TRANSACTIONAL SOURCE OF TRUTH	MATHEMATICAL VARIANCE / ROOT DISCREPANCY	VERDICT	DEFECT
01	Profit & Loss <code>/admin/reports/profit-loss</code>	Sales: ৳ 151,000 COGS: (৳ 120,400) Gross: ৳ 30,600 Expenses: (৳ 0) Net: ৳ 30,600	Cash Flow Paid Expenses = ৳ 6,300 Ad Spend = ৳ 700 Asset Maintenance = ৳ 1,000	Expenses understated by ৳ 6,300.00. Net Profit is artificially inflated by ৳ 6,300. Net margin falsified at 20.26% vs actual 16.09%.	CRIT FAIL	BUG-114
02	Supplier Payments <code>/admin/reports/supplier-payments</code>	Purchases: ৳ 178,000 All-Time Paid: ৳ 178,100 Period Paid: ৳ 193,100	Purchase Report Bills = ৳ 183,000 Cash Flow Supplier Paid = ৳ 183,000 Supplier Advance = ৳ 10,100	Period Paid exceeds All-Time Paid by ৳ 15,000. Purchases understated by ৳ 5,000. All-Time query misses advances while Period query includes them.	CRIT FAIL	BUG-109
03	Cash Movement <code>/admin/reports/cash-movement</code>	Inflow: ৳ 173,000 Outflow: ৳ 191,600 Balance: -৳ 18,600 (Deficit)	Cash Flow In = ৳ 369,500 Cash Flow Out = ৳ 315,400 Net Liquidity = +৳ 54,100	Reports false deficit of -৳ 18,600. Query omits Capital In (৳ 100k), Loans (৳ 20k), PR Refunds (৳ 29k). Exposes raw string <code>sale_return_refund</code> .	CRIT FAIL	BUG-111
04	Sales Summary <code>/admin/reports/sales</code>	Total Sales: ৳ 125,500 Collected: ৳ 136,000 Due: ৳ 15,000	Detail Sales = ৳ 151,000 Sale Returns = ৳ 25,500 Gross: ৳ 136k + ৳ 15k = ৳ 151k	Total Sales (৳ 125.5k) < Collected (৳ 136k). Query silently deducts returns from sales column while leaving collections at gross without a "Returns" column.	FAIL	BUG-118

#	REPORT MODULE & PATH	REPORTED LIVE VALUES	TRANSACTIONAL SOURCE OF TRUTH	MATHEMATICAL VARIANCE / ROOT DISCREPANCY	VERDICT	DEFECT
05	Monthly Summary /admin/reports/monthly-summary	Sept Sales: ₺ 125,500 Sept Paid: ₺ 136,000 Sept Due: ₺ 15,000	Detail Sales = ₺ 151,000 Sale Returns = ₺ 25,500	Inherits identical sales deduction paradox where Total Sales is less than Total Paid. Navigation labeled "Monthly Summary" inside yearly view.	FAIL	BUG-119
06	Receivables Aging /admin/reports/receivables-aging	Customer: Chaim Bowen Due: ₺ 15,000 Days: 2.0033220333565	Sales Due = ₺ 15,000 Invoice Date: 06 Sep 2026 Current Date: 08 Sep 2026	Overdue balance matches, but Days Overdue displays raw floating point timestamp subtraction instead of integer formatting.	PARTIAL	BUG-121
07	Daily Summary (DTS) /admin/reports/dts	Current Date: ₺ 0 06 Sep: Sales ₺ 125k, Purch ₺ 140k 07 Sep: Sales ₺ 500, Purch ₺ 13k	Purchases match Purchase Report exactly (₺ 140k & ₺ 13k). Sales reflect net after returns (₺ 125k & ₺ 500).	Defaults to ₺ 0 with no fallback when today has no sales. Replicates return deduction mismatch without separate refund card.	PARTIAL	BUG-102
08	Category-wise Sales /admin/reports/category-wise	Category: "Shirt" Sold: 8 items Rev: ₺ 151,000 Profit: ₺ 30,600	6 x Mobile (₺ 150,000) 2 x Full Shirt (₺ 1,000) Total: ₺ 151,000	Math is internally consistent with Detail Sales, but mobile devices are erroneously grouped under the "Shirt" category.	PARTIAL	Catalog Defect
09	Detail Sales /admin/reports/detail-sales	6 Invoices (8 line items) Grand Total: ₺ 151,000	Sales Invoices S001-S004 (06 Sep), S001-S002 (07 Sep)	100% Reconciliation. Correct item prices, quantities, and line totals across all invoices.	PASS	Baseline OK
10	Purchase Report /admin/reports/purchase	9 Bills (4 on 07 Sep, 3 on 06 Sep, 2 in Aug) Total Purchases: ₺ 183,000 Paid: ₺ 183,000	Cash Flow "Supplier Payment" = ₺ 183,000 Purchases List Total = ₺ 183,000	100% Reconciliation. Exactly matches Cash Flow supplier disbursements and Purchase Order register.	PASS	Baseline OK
11	Inventory Report /admin/reports/inventory	Full Shirt: 259 in stock Stock Value: ₺ 51,800 Mobile: 0 (Out of stock)	Initial + Purchases – Returns – Sales = 259 shirts. Unit Cost = ₺ 200.00	100% Reconciliation. 259 × ₺ 200 = ₺ 51,800.00. Valuation and inventory balances match perfectly.	PASS	Baseline OK
12	Customer Report /admin/reports/customer	Chaim Bowen: Spent ₺ 25k, Due ₺ 15k Rhannon: Spent ₺ 500, Due ₺ 0	Sales Invoices S20260906002 and S20260907002	100% Reconciliation. Outstanding receivables match Sales Report Due (₺ 15,000.00).	PASS	Baseline OK
13	Staff Report /admin/reports/staff	EMP-0001: Present 2, Absent 2, Late 1, 27.0h worked	Attendance & Time Clock Register	100% Reconciliation. Accurately logs operational hours and attendance statistics.	PASS	Baseline OK

3. Deep-Dive: Reconciliation by Transaction Stream

Detailed double-entry validation tracking transactions from operational entry modules through Central Cash Flow into final financial statements.

3.1. Operating Expenses vs Cash Flow vs P&L (The ₺ 6,300 Leak)

- Source Ledger (/admin/expenses):** Verified 3 Paid expense vouchers totaling ₺ **6,300.00** (EXP-2026-0001 : ₺ 5,000; EXP-2026-0005 : ₺ 300; EXP-2026-0007 : ₺ 1,000). Pending/Unapproved expenses (EXP-0002 , 0003 , 0004 , 0006) total ₺ 670.00.
- Cash Flow Reconciliation:** The Cash Flow statement accurately recognizes only the **Paid** expenses as ₺ **6,300.00** under Operating Cash Outflows.
- The Critical Defect:** The **Profit & Loss Report** outputs `Less: Operating Expenses (₺ 0)`. Net Profit is reported at ₺ **30,600.00**. Correct Net Profit must be:

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Correct Net Profit = Gross Profit (₺ 30,600.00) - Operating Expenses (₺ 6,300.00) = ₺ 24,300.00
Reported Net Margin: 20.26% (INFLATED) | Actual Net Margin: 16.09%
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3.2. Fixed Assets & Purchases vs Cash Flow (100% Parity)

- Asset Register (/admin/assets):** Contains AST-0001 (Office printer, ₺ 3,000) and AST-0002 (Tissue dispenser, ₺ 300). Total capitalized assets = ₺ **3,300.00**.
- Cash Flow Reconciliation:** Cash Outflow line item **"Asset Purchase"** = ₺ **3,300.00** (100% Match).
- Maintenance:** Cash Outflow line item **"Asset Maintenance"** = ₺ **1,000.00** (100% Match with maintenance log).

3.3. Loans & Liabilities vs Cash Flow (100% Parity)

- Commercial Term Loans (/admin/loans):** Lender Camille Walls — Principal ₺ 20,000.00; Repaid ₺ 1,666.67.
Cash Flow: `Loan Received (Lender)` = ₺ **20,000.00** (Inflow); `Loan Repayment (Lender)` = ₺ **1,666.67** (Outflow). (100% Match)
- Personal Loans (/admin/personal-loans):** Orla Calderon: Taken ₺ 10,000, Repaid ₺ 4,000. Maya Rhodes: Given ₺ 6,000, Recovered ₺ 3,000.
Cash Flow: `Loan Taken (Person)` = ₺ **10,000.00**; `Loan Paid Back` = ₺ **4,000.00**; `Loan Given` = ₺ **6,000.00**; `Loan Recovered` = ₺ **3,000.00**. (100% Match)

3.4. Purchases & Supplier Payments (Paradox Explained)

- Purchases (/admin/purchases):** 9 POs totaling ₺ **183,000.00** (all Paid). Matches Cash Flow "Supplier Payment" = ₺ **183,000.00**.
- Advances:** Supplier Advance = ₺ **10,100.00**; Advance Refund = ₺ **3,000.00**.
- The Paradox in /admin/reports/supplier-payments :** Shows `All-Time Paid: ₺ 178,100.00` while `Period Paid: ₺ 193,100.00`.
Root Cause: The period query aggregates purchase payments (₺ 183k) + advances (₺ 10.1k) = ₺ 193,100, but the all-time query joins purchases directly, omitting advance vouchers. In double-entry accounting, Period Paid cannot exceed Lifetime Paid.

3.5. Sales & Customer Dues (The Return Deduction Mismatch)

- Gross Invoiced Sales:** 6 invoices across 06 Sep and 07 Sep totaling ₺ **151,000.00**.
- Customer Returns:** SR-2026-0001 (₺ 25,000) and SR-2026-0002 (₺ 500) = ₺ **25,500.00**.
- Sales Report Mismatch:** In /admin/reports/sales , the column header states **Total Sales** = ₺ **125,500** while Collected = ₺ **136,000** and Due = ₺ **15,000**. Because ₺ 136k + ₺ 15k = ₺ 151k, calling ₺ 125.5k "Total Sales" creates an accounting contradiction. The column must be renamed "Net Sales", or an explicit "Returns" column added.

4. Updates Committed & Pushed to Remote Repository

All audit deliverables, standalone forensic defect files, updated RTM, and test automation fixes have been permanently committed to GitHub:

GIT REPOSITORY STATUS:

Remote URL: <https://github.com/naimuropu-cell/bizpos-qa-audit-framework.git>

Active Branch: `main` | Latest Commit: `791d956` | Status: Up to date with origin/main

COMMITTED FILE PATH	TYPE / SCOPE	AUDIT SIGNIFICANCE & REMEDIATION GUIDANCE FOR DEV TEAM
04-bug-reports/ALL_DEFECTS_LOG.md	CATALOG	Upgraded from 99 to 118 audited defects (91 Fixed, 6 Need Revisions - QA, 18 Not Fixed, 3 Invalid). Full reference catalog for sprint planning.
Bug-114_Profit_Loss_Zero_Expense.md	DEFECT SPEC	Forensic proof and remediation patch for the P&L ₪ 0 expense leak. Includes exact SQL fix binding <code>total_expenses</code> to Paid expenses.
Bug-109_Supplier_Payments_Paradox.md	DEFECT SPEC	Detailed explanation of the Period Paid (₪ 193.1k) vs All-Time Paid (₪ 178.1k) calculation paradox with unified SQL query.
Bug-111_Cash_Movement_Deficit.md	DEFECT SPEC	Step-by-step fix to eliminate the false -₪ 18,600 deficit by including capital, loan, and return inflows in the Cash Movement controller.
Bug-118_Sales_Report_Paradox.md	DEFECT SPEC	Recommends layout change replacing confusing "Total Sales" with explicit "Gross Sales", "Returns", and "Net Sales" columns.
03-traceability-matrix/RTM_BizPOS.md	RTM	Added requirements <code>REQ-REP-01</code> to <code>REQ-REP-05</code> mapping all reporting business rules directly to test cases and defects.
playwright-tests/pages/ & fixtures/	AUTOMATION	Updated Page Object Models with verified live production DOM selectors (<code>#loginEmail</code> , <code>#loginPassword</code> , <code>#loginBtn</code>) and <code>/admin/money/cashflow</code> .